Proposal presentation script

Qihang Feng | ECE 730 | About five minutes

Slide 1 Effective prediction utilization

Hi, everyone. My project will explore how to make better use of a robot control model's own predictions. I'll use a two-dimensional simulator where a pusher moves a T-shaped block onto a target. Visual feedback connects physical motion with software decisions.

Slide 2 Prediction and execution

Our baseline uses receding-horizon control. It executes fixed eight movements, then discards the remaining predictions and replans. This is a standard design choice.

However, some unused predictions may still be useful. So I'm going to implement a controller that uses several feedback signals to decide whether to replan or continue executing the remaining predicted movements.

Slide 3 Signals for continuing or replanning

Specifically, the controller will check current target alignment, recent alignment changes, and the smoothness of the remaining movements.

For example, If the visual estimate is clear, alignment loss is limited, and the remaining movements are smooth, then it can just continue up to twelve movements. Otherwise, it will just replan. Also, when the block is near the target, it will also replan after the first eight movements.

Additionally, repeated warnings and accumulated alignment loss may also trigger an earlier interruption. But a temporary alignment drop alone will not, because it could be part of a useful adjustment. Especially when the object near the target.

Slide 4 Self imitation based fine tuning

The second approach is self-imitation-based fine-tuning. Basically, I'm going to use the model's own successful movements to improve the model itself.

Specifically, I will select intermediate states from failed baseline attempts. Then, I will restart from these states and let the original model try again several times. If a retry eventually completes the task, I will keep its successful recovery trajectory.

After that, I will combine these trajectories with expert demonstrations to fine-tune only the model's final output block. The other parameters will remain fixed. Finally, the updated model will use the same adaptive controller.

Slide 5 Planned evaluation

As for the evaluation, I will compare four configurations in an ablation study to understand what adaptive execution and fine-tuning contribute, both separately and together.

The metrics will include success rate, average final alignment, average prediction calls, and average runtime.

Slide 6 Expected contribution and scope

The target outcome is a higher success rate, with fewer predictions and shorter runtime on average.

I will also report average final alignment, but improving it is not my main objective. The main focus is helping the robot complete more tasks successfully.

Slide 7 Thank you

That's all. Thank you for listening.